ECRS 2026 Conference
The 6th International Electronic Conference on Remote Sensing
19–21 October 2026 | Online
MDPI

Twelve-Year Landsat-Derived Surface Urban Heat Island Intensity by Local Climate Zone in Barcelona (2013–2024)

Kerin Cardona*, Jordi Cipriano, Gerard Mor
International Centre for Numerical Methods in Engineering (CIMNE), Building Energy and Environment Group, Lleida, Spain
*Corresponding author: kcardona@cimne.upc.edu

INTRODUCTION & AIM

Outdoor thermal exposure depends strongly on urban surface temperature, yet long-term, high-resolution land surface temperature (LST) climatologies remain scarce for Mediterranean cities:

- **Weather stations:** point air temperature only
- **ERA5-Land:** skin temperature at ~9 km, no intra-urban structure
- **Single satellite scenes:** snapshots that depend on the weather of the day

Aim: a 12-year, 30 m LST climatology for Barcelona, read through Local Climate Zones (LCZ) and turned into hotspot, building and census-section exposure indicators for adaptation planning.

METHOD

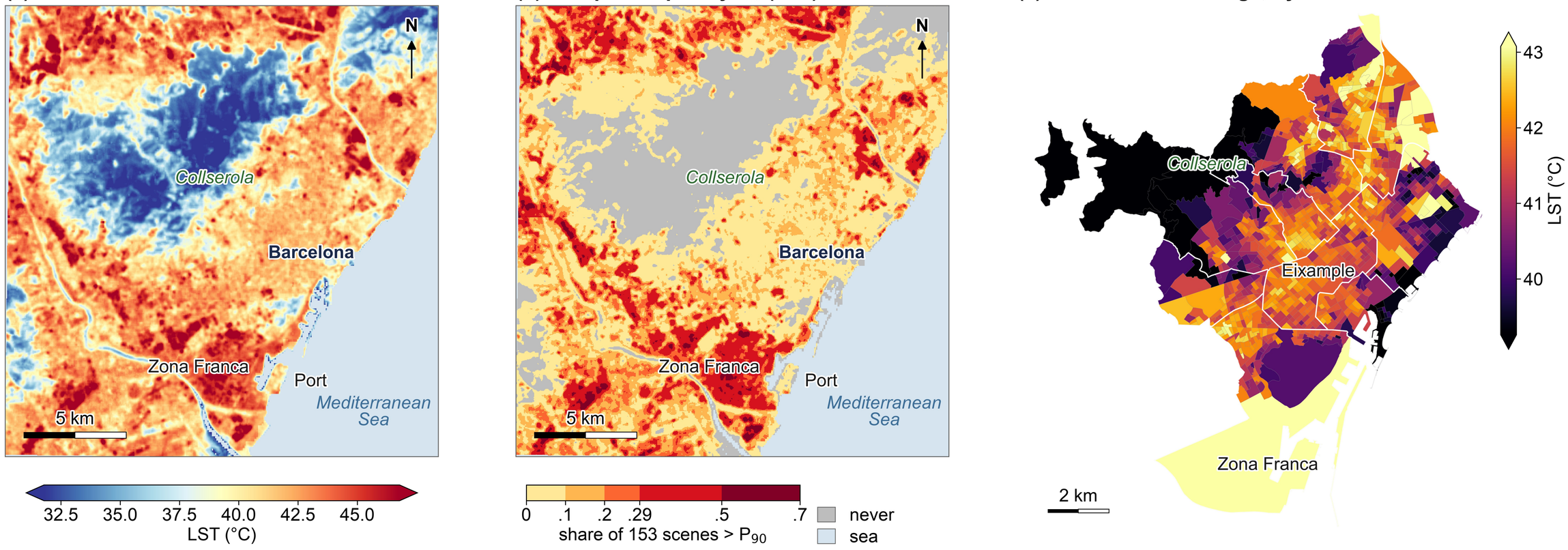
Barcelona metropolitan core: 464 km² window (~21 × 22 km), 13 municipalities.

- **378** cloud-filtered Landsat 8/9 Collection 2 Level-2 scenes, 2013–2024 (153 in summer), ~10:30 local; 30 m grid, ~100 m native thermal
- **Climatology:** seasonal median, MAD and clear-sky count per pixel
- **SUHI by LCZ** (Bechtel et al. 2019): class median minus LCZ D (low plants, 315.0 K)
- **Hotspot frequency:** share of summer scenes above the 90th percentile of the JJA field (317.2 K)
- **Drivers:** gradient boosting + SHAP, 13 covariates, 37,345 land cells, 2 km spatial-block cross-validation
- **Cross-checks:** ERA5-Land (325 pairs); ECOSTRESS 70 m (7 scenes, $|\Delta t| \leq 2$ h)

RESULTS & DISCUSSION

Summer (JJA) daytime, 12-year climatology. 106,203 buildings and 1,068 census sections linked to it.

(a) JJA median LST, 2013–2024 | **(b) Hotspot frequency f_{hs} (JJA)** | **(c) LST around buildings, by census section**



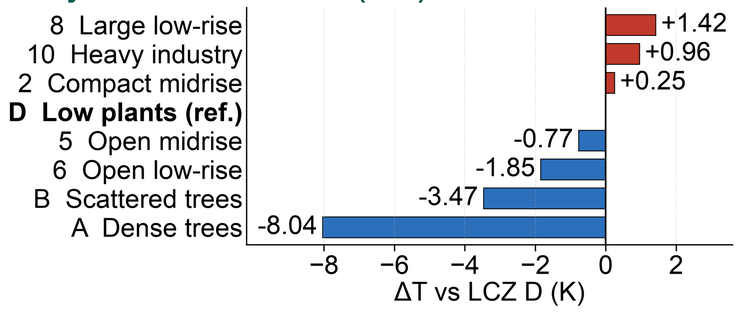
(a) The Llobregat industrial corridor and dense cores exceed 45 °C; Collserola forest is the main cool feature. **(b)** 64 % of the window (78 % of land) is a hotspot at least once, 1.1 % (1.3 %) in over half of summer scenes. **(c)** Median LST in 30 m building buffers, 1,068 census sections of Barcelona city; port and logistics areas are the hottest.

Cross-checks of the 30 m field

Reference	n	Bias (K)	r
ERA5-Land, all	325	−4.32	0.92
ERA5-Land, JJA	134	−5.57	0.26
ECOSTRESS, pooled	7	+0.37	0.92
ECOSTRESS, per scene	7	−6.4…+4.3	0.43–0.96

Bias = reference − Landsat; n = matched Landsat scenes. ERA5 follows the seasons but not the city. ECOSTRESS places warm and cool areas in the same locations; n = 7 cannot establish absolute accuracy.

SUHI by Local Climate Zone (JJA)



Exposure indicators

- 106,203 buildings, 13 municipalities: LST in 30 m and 100 m buffers (r = 0.989)
- Only 10,333 distinct values: Landsat resolves ~100 m context, not single buildings
- 1,068 census sections: median 34.9–47.2 °C, but IQR only 41.1–42.1 °C; extremes set by land use (port vs seafront, parks)

CONCLUSIONS

- A tree-canopy cool island (−8.0 K; still 8.4 K below LCZ 8 at equal elevation) dominates over a weak built heat island (≤ +1.4 K).
- Vegetation (NDVI) is the leading driver (held-out $R^2 = 0.84$; ranked first in 5/5 splits): tree cover is the main modifiable cooling lever.
- Reproducible 30 m baseline from open data for adaptation planning; ERA5-Land cannot replace it.

FUTURE WORK/ REFERENCES/ACKNOWLEDGMENT

Future work: combine exposure with socio-economic vulnerability; year-split validation; night-time LST.

Data: Landsat 8/9 C2 L2, ECOSTRESS L2G v002, ERA5-Land, INSPIRE cadastre, Copernicus DEM / HRL / Urban Atlas; LCZ map: Demuzere et al. (2019).

Session S5 — Remote Sensing in Urban Areas